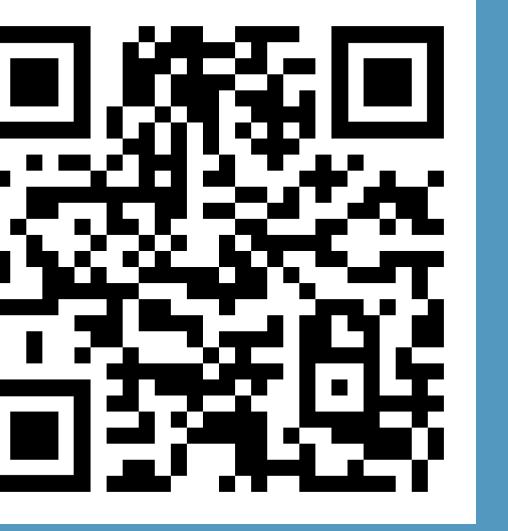




Equivariant Robot Learning with Images

Key Insight. By expressing RGB images as collections of 3D rays rather than 2D pixels, we enable the first end-to-end SE(3)-equivariant robot policy that operates on raw RGB cameras—no point clouds or depth sensors required.

Equivariance. A policy f is equivariant to a symmetry group G if:

$$\rho'_g f(x) = f(\rho_g x) \quad \forall g \in G$$

The Challenge. A 2D image has no inherent 3D geometry—pixels transform under planar SE(2) only. Prior equivariant methods require point clouds or structured top-down views, preventing use with standard RGB setups like ALOHA or UMI.

Representing Images as Rays in 3D

Given a calibrated camera with extrinsics $M = [R \mid \mathbf{t}] \in \text{SE}(3)$ and intrinsics $K \in \mathbb{R}^{3 \times 3}$, each pixel $\mathbf{u} \in \mathbb{Z}^2$ maps to a unique 3D ray:

$$\mathbf{r} = (\mathbf{t}, RK^{-1}\mathbf{u})$$

where $\mathbf{r} = (\mathbf{r}_o, \mathbf{r}_d)$ describes the ray's *origin* $\mathbf{r}_o \in \mathbb{R}^3$ and *direction* $\mathbf{r}_d \in S^2$. Images from multiple arbitrarily-placed cameras are unified in a single 3D world frame.

Assumption. Camera intrinsics and extrinsics are known. Our method is robust to calibration noise—only 1% average drop despite errors at test time.

SE(3) Geometric Tokens

Each image patch is paired with a 3D pose describing where it sits in the world (camera position) and how it is oriented (ray direction = z , image-plane axes = x, y). This pairing of *visual features* and *3D pose* is a **geometric token**.

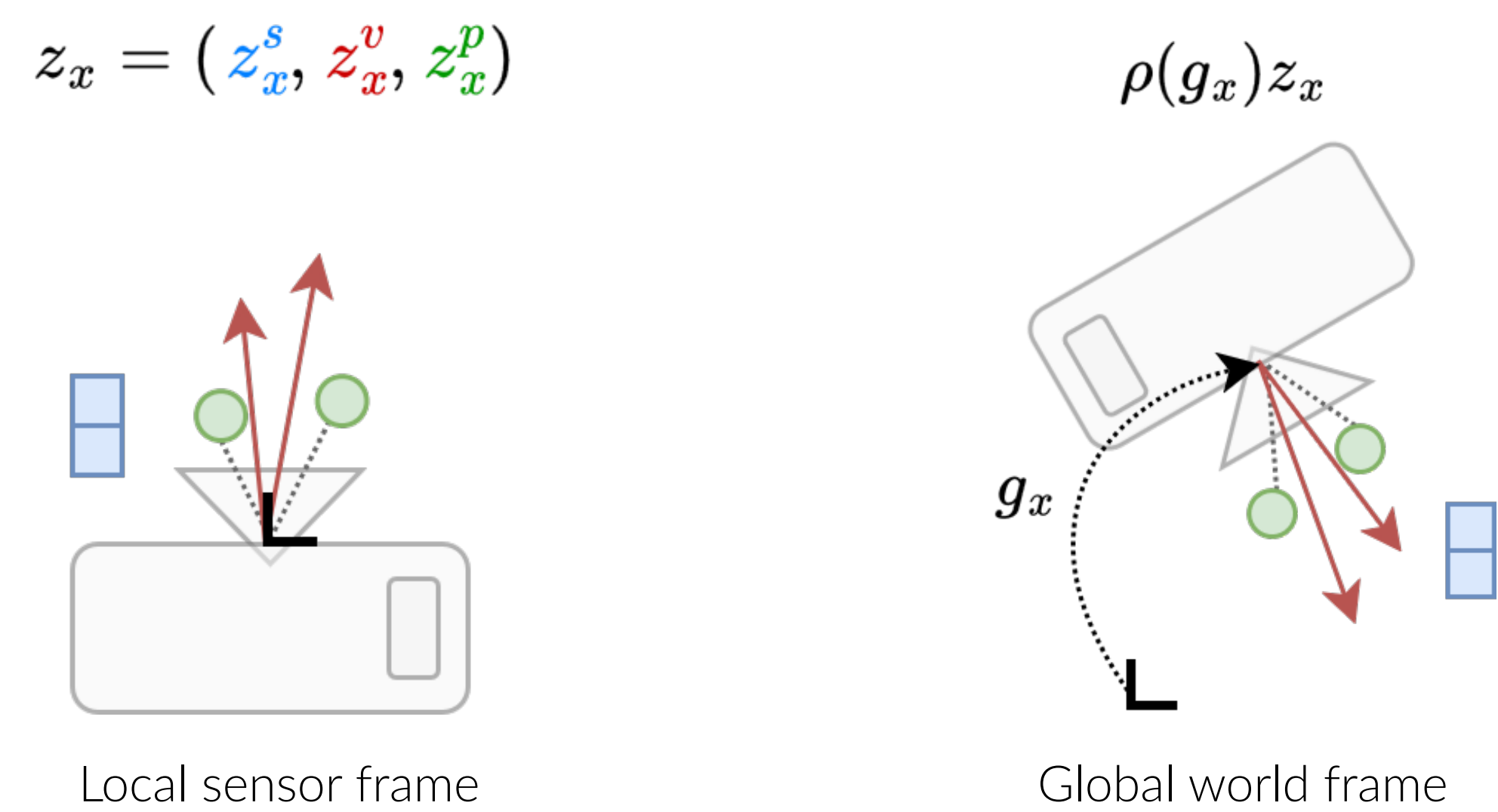


Figure 1. Features are stored in the camera's local frame (left); the token's SE(3) pose g_x transforms them into the shared world frame (right).

Formally, a token $x = (z_x, g_x)$ stores features z_x in the camera's local frame and uses pose $g_x \in \text{SE}(3)$ to place them in the world:

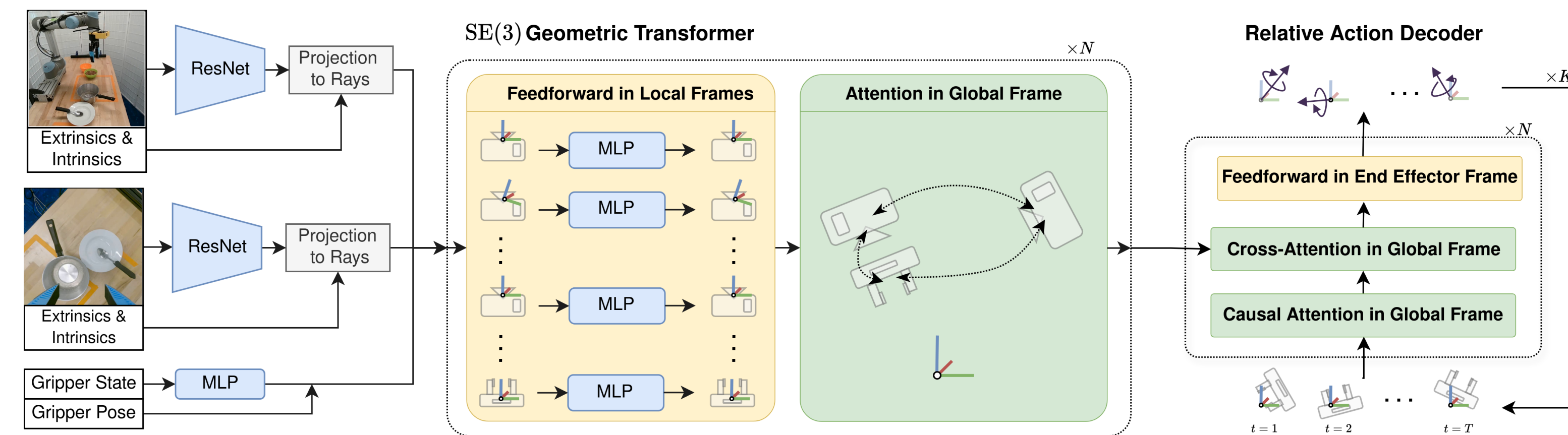
$$\rho(g_x) z_x = \left(\underbrace{\rho^s(g_x) z_x^s}_{\text{scalar (invariant)}}, \underbrace{\rho^v(g_x) z_x^v}_{\text{vector (rotate)}}, \underbrace{\rho^p(g_x) z_x^p}_{\text{point (rotate+translate)}} \right)$$

Because all tokens share a common world frame, features from multiple cameras can be directly compared and fused.

Other Modalities. Robot proprioception (end-effector pose, joint states) is also encoded as geometric tokens: features based on joint states associated with the end-effector's 3D pose. This gives the model a unified representation across all input modalities.

The first equivariant robot policy for any RGB camera setup, robust to new viewpoints without retraining

RAVEN Architecture

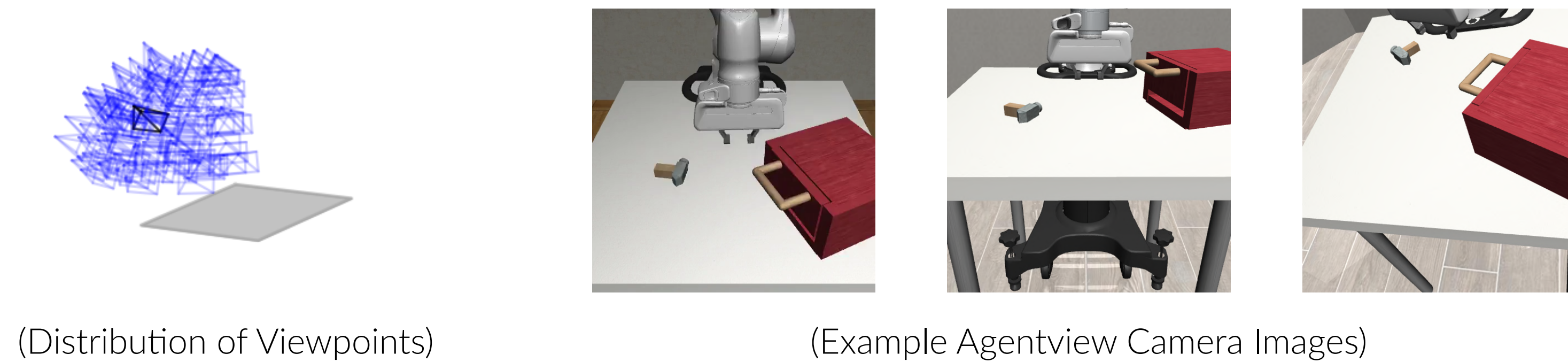


- Image Encoder:** A pretrained ResNet extracts patch-level features. Camera intrinsics and extrinsics assign each patch an SE(3) pose, forming geometric tokens.
- SE(3) Geometric Transformer:** Geometric Transform Attention (GTA) layers perform cross-token attention in the global frame while computing features in each token's local frame, fusing all cameras and proprioception equivariantly.
- Flow Matching Decoder:** Action tokens anchored at the current end-effector pose are iteratively refined by a cross-attention decoder trained with a flow matching objective to predict SE(3) gripper trajectories.

Equivariance Properties. RAVEN has guaranteed generalization to global SE(3) transformations of the robot, sensors and environment. This is slightly weaker than equivariance claims of point cloud methods, due to occlusion.

Generalization to Novel Camera Viewpoints

We test performance when camera viewpoint is randomized at start of each episode ($\pm 20^\circ$ pitch, $\pm 40^\circ$ yaw).



Method	Avg	Stack	Stack 3	Coffee	Cleanup
RAVEN	71	99	68	53	65
EquiDiffPo	48	65	25	51	52
DiffPo (Pre)	44	78	42	21	36

Table 1. Success rate (%) on MimicGen with randomly positioned agentview camera.

Takeaway. RAVEN gets **+23%** over the best baseline on unseen viewpoints. Compared to a fixed agentview camera, RAVEN's performance drops only 8% while EquiDiffPo drops 20%.

Note. RAVEN is not equivariant to individual camera viewpoint transformations, but the geometric structure of its ray-based features enables superior generalization in this setting.

Simulation Experiments

We evaluate on **12 MimicGen** single-arm tasks (100 demos; agent-view + eye-in-hand cameras) and **6 DexMimicGen** bimanual tasks (50 demos; three cameras including two moving eye-in-hand cameras).



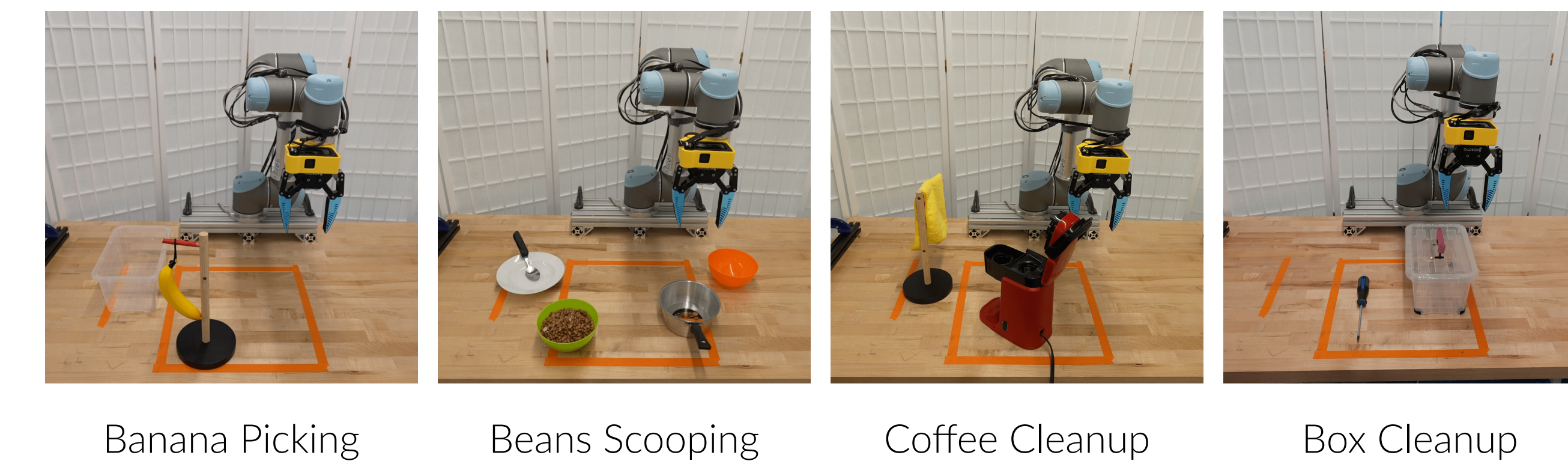
Method	MimicGen	DexMimicGen
RAVEN	66 (+12)	82 (+17)
EquiDiffPo	54	62
DiffPo (Pre)	52	65
DiffPo	42	60

Table 2. Success rate (%) averaged across tasks. (Pre) means pretrained image encoder.

These benchmarks show that RAVEN seamlessly handles varying numbers of cameras and end-effector configurations with no architectural changes.

Real World Experiments

We evaluate RAVEN on a physical robot arm where real-world conditions introduce more complex visual features, and imperfect camera calibration.



Method	Avg.	Banana	Beans	Coffee	Box
RAVEN	63	100	45	45	60
DiffPo (Pre)	24	50	0	15	30

Table 3. Success rate (%) over 20 trials. Policies trained on 70 demos (45 for box cleanup).

Takeaway. +39% success rate and +35% task progress over DiffPo (Pre).

Failure modes: Policy gets stuck repeating same motions, likely due to limited observation history.

Conclusion

RAVEN combines the data efficiency of equivariant methods with the flexibility of standard RGB-based policies. Unlike prior equivariant methods, it works with any RGB camera, benefits from pretrained image encoders, and trains faster than non-equivariant diffusion baselines.

Contributions:

- SE(3) Geometric Tokens:** a unified equivariant representation from any number of arbitrarily placed RGB cameras
- RAVEN:** first end-to-end SE(3)-equivariant RGB policy; +12% (MimicGen), +35% (real world), 1.6× faster training